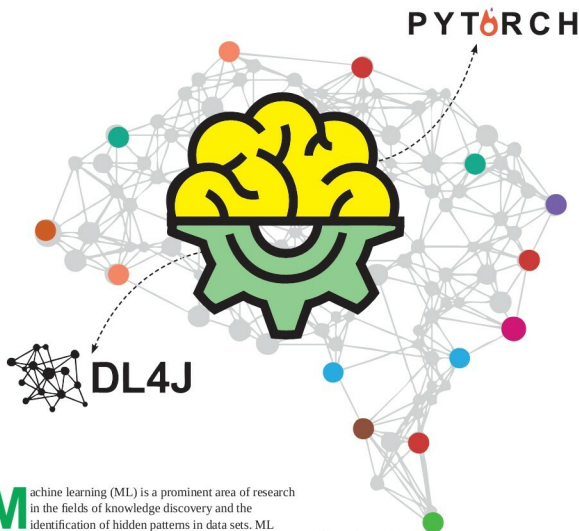


DeepLearning4j and PyTorch: Two Powerful Deep Learning Tools

This article gives an introduction to two free and open source tools for deep learning and knowledge discovery—DL4J and PyTorch.



Machine learning (ML) is a prominent area of research in the fields of knowledge discovery and the identification of hidden patterns in data sets. ML integrates predictive mining and the optimisation of results in multiple applications including bioinformatics, biometrics, computer vision, computational anatomy, criminal forensics, face detection, document forgery, fraud detection and many others. A comprehensive list of such applications is available on the Internet. ML focuses on the training of intelligent models so that the prediction can be accurate and fast.

Machines learn from the data and then get trained for prediction. For example, the preferences of customers are fetched from their existing records of interests.

Deep learning

Deep learning (DL) is a more accurate and performance-aware flavour of machine learning, in which the error rate is very small compared to the latter. DL is also known as deep structured learning and hierarchical learning.

Let's look at speech recognition, for instance. If we install speech recognition software in a mobile phone for disaster/emergency purposes, when selected words like 'help' or 'kidnap' are uttered, a disaster message will be sent to chosen family members or friends. In this case, ML